Assessment

Bookmark | Full Screen

Week 2 : Assignment 2

Your last recorded submission was on 2026-08-04, 10:45 IST

Due date on 2026-08-05, 23:59 IST

Thank you for taking the Week 2 : Assignment 2

1. Given a training data set of 10,000 instances, with each input instance having 17 dimensions and each output instance having 2 dimensions, the dimensions of the design matrix used in applying linear regression to this data is — 1 Point

- ○ 10000 × 17
- ◉ 10002 × 17
- ○ 10000 × 18
- ○ 10000 × 19

2. Suppose we want to add a regularizer to the linear regression loss function, to control the magnitudes of the weights β. We have a choice between $\Omega_1(\beta) = \Sigma_{i=1}^{p}|\beta|$ and $\Omega_2(\beta) = \Sigma_{i=1}^{p}\beta^2$ Which one is more likely to result in sparse weights? — 1 Point

- ◉ Ω_1
- ○ Ω_2
- ○ Both Ω_1 and Ω_2 will result in sparse weights
- ○ Neither of Ω_1 or Ω_2 can result in sparse weights

3. The model obtained by applying linear regression on the identified subset of features may differ from the model obtained at the end of the process of identifying the subset during — 1 Point

- ○ Forward stepwise selection
- ○ Backward stepwise selection
- ◉ Forward stagewise selection
- ○ All of the above



4. Consider forward selection, backward selection and best subset selection with respect to the same data set. Which of the following is true? — 1 Point

- ◉ Best subset selection can be computationally more expensive than forward selection
- ○ Forward selection and backward selection always lead to the same result
- ○ Best subset selection can be computationally less expensive than backward selection
- ○ Best subset selection and forward selection are computationally equally expensive
- ○ Both (b) and (d)

5. In the lecture on Multivariate Regression, you learn about using orthogonalization iteratively to obtain regression co-effecients. This method is generally referred to as Multiple Regression using Successive Orthogonalization. In the formulation of the method, we observe that in iteration k, we regress the entire dataset on $z_0, z_1, \ldots z_{k-1}$. It seems like a waste of computation to recompute the coefficients for z_0 a total of p times, z_1 a total of $p-1$ times and so on. Can we re-use the coefficients computed in iteration j for iteration $j+1$ for z_{j-1} ? 1 Point

- ○ No. Doing so will result in the wrong γ matrix. and hence, the wrong β_i 's.
- ◉ Yes. Since z_{j-1} is orthogonal to $z_{j-l} \forall l \leq j1$, the multiple regression in each iteration is essentially a univariate regression on each of the previous residuals. Since the regression coefficients for the previous residuals don't change over iterations, we can re-use the coefficients for further iterations.

6. Principal Component Regression (PCR) is an approach to find an orthogonal set of basis vectors which can then be used to reduce the dimension of the input. Which of the following matrices contains the principal component directions as its columns (follow notation from the lecture video) 1 Point

- ○ X
- ○ S
- ○ X_c
- ◉ V
- ○ U

7. Consider the following five training examples 2 Point

x	y
2	0.8978
3	12.7586
4	16.3192
5	19.3129
6	21.1351

We want to learn a function $f(x)$ of the form $f(x) = ax + b$ which is parameterised by (a, b). Using squared error as the loss function, which of the following parameters would you use to model this function to get a solution with the minimum loss.

- ○ (4,3)
- ○ (1,4)
- ○ (4,1)
- ◉ (3,4)

8. Here is a data set of words in two languages. 2 Point

Word	Language
piano	English
cat	English
lepto	Vinglish
shairo	Vinglish

Let us build a nearest neighbours classifier that will predict which language a word belongs to. Say we represent each word using the following features.
- Length of the word
- Number of consonants in the word
- Whether it ends with the letter 'o' (1 if it does, 0 if it doesn't)

For example, the representation of the word 'waffle' would be [6, 2, 0]. For a distance function, use the Manhattan distance.

$d(a, b) = \Sigma_{i=1}^{n} |a_i - b_i|$ where $a, b \in R^n$

Take the input word 'keto'. With k = 1, the predicted language for the word is?

- ◉ English
- ○ Vinglish
- ○ None of the above